

Acharya Kannada Vedike (AKV) — Nuditaranga 2026

Comprehensive System Architecture, Technology Stack & Operational Manual

Version: 2.0.0 (Production Stable)	Hosting: Vercel Edge + Serverless	Database: Supabase PostgreSQL (Mumbai)
Live URL: akv-nuditaranga-2026.vercel.app	Runtime: Python 3.12 / Node 24	Security: Bcrypt + HS256 JWT RBAC

1. Executive Summary & Vision

Acharya Kannada Vedike (AKV) is the official cultural association of Acharya Institutes, Bengaluru, dedicated to the preservation, promotion, and celebration of Karnataka's rich linguistic, artistic, and literary heritage. **Nuditaranga 2026 (■■■■■■■■■■ ■■■■)** is its premier inter-collegiate annual festival, hosting thousands of students across traditional arts, literary contests, musical performances, and theatrical showcases. This platform is an enterprise-grade, cloud-native web system providing end-to-end automation for fest enrollment, instant digital QR pass issuance, mobile camera admission verification, volunteer shift tracking, and executive analytics.

2. Target Cloud & Serverless Architecture

The system employs a high-performance, decoupled client-server architecture deployed on **Vercel Edge & Serverless Functions** backed by a managed **Supabase PostgreSQL** relational database:

Tier	Component	Implementation & Architectural Role
Client Layer	Vite + React 18 SPA	Progressive web client with Glassmorphic design, dual Kannada/English language support, dynamic routing, and offline localStorage caching.
Edge CDN	Vercel Edge Network	Globally distributed edge caching for static assets, automatic SSL termination, and unified /api/* rewrite routing.
API Gateway	Vercel Serverless Function	Python 3.12 ASGI runtime (api/index.py) executing FastAPI endpoints with zero server maintenance and automatic horizontal scale-out.
Backend	FastAPI Framework	High-throughput async API handling user auth, registration workflows, role-based access control, QR token verification, and data exports.
ORM Layer	SQLAlchemy 2.0	Robust object-relational mapping configured with pool_pre_ping=True and pool_recycle=300 to eliminate serverless stale connections.
Database	Supabase PostgreSQL	Hosted relational database (AWS Mumbai ap-south-1) accessed via dedicated PgBouncer Transaction Pooler (Port 6543) for persistent storage.
Dev DB	SQLite (Local Only)	Local isolated database (akv_fest.db) strictly restricted to local machines. Production guards actively prevent ephemeral SQLite execution.

3. Detailed Technology Stack & Libraries

Domain	Technology / Tool	Version	Strategic Justification & Usage
Frontend UI	React.js	18.3.1	Declarative component architecture power dynamic forms, modals, and tabbed dashboards.
Build Engine	Vite	8.3.0	Sub-second HMR and optimized Rollup production asset chunking with built-in API proxy.
Iconography	Lucide React	0.468.0	Modern, feather-weight SVG icon set for clean and responsive touch-friendly navigation.
QR Engine	html5-qrcode	2.3.8	Direct hardware camera interface for rapid attendee QR check-in on mobile browsers.
Pass QR	qrcode.react	4.1.0	Vector SVG QR code generation for student digital entry badges.
Celebration	canvas-confetti	1.9.4	Delightful particle animation triggered on successful registration and check-in.
Backend API	FastAPI	0.141.1	Modern Python ASGI framework offering native async execution and automatic OpenAPI specs.
Validation	Pydantic	2.13.5	Strict input schemas validating AUID formats, college emails, phone numbers, and roles.
ORM	SQLAlchemy	2.0.54	Database-agnostic relational abstraction with robust foreign keys, cascades, and pooling.
Postgres Driver	psycopg2-binary	2.9.13	High-speed C-based PostgreSQL database adapter for Python.
Auth & Crypt	Bcrypt + PyJWT	5.0 / 2.15	Industry-standard salt-and-stretch password hashing with stateless HS256 JWT tokens.
Reporting	OpenPyXL	3.1.5	Automated server-side generation of styled Excel attendance and registration spreadsheets.

4. Relational Database Schema & Entities

The relational schema is normalized into 10 structured entities enforcing relational integrity, cascade deletions on dependent tokens, and indexed lookups:

Table Name	Primary Key	Foreign Keys	Indexed & Unique Attributes	Functional Description
users	id (Serial)	None	aid (UQ), email (UQ), registration_id (UQ), role	Stores students, volunteers, and admin user profiles with bcrypt password hashes.
admins	id (Serial)	user_id -> users.id	username (UQ), approval_status	Coordinator profiles requiring Superadmin approval (PENDING, APPROVED, REJECTED).
events	id (String)	None	id, category	Fest competitions with dual-language titles, rules, slots, venue, and team parameters.
registrations	id (Serial)	event_id -> events.id user_id -> users.id	registration_id (UQ), usn, aid	Event enrollment records containing team rosters, checkin status, and timestamps.
volunteer_attendance	id (Serial)	user_id -> users.id	aid, date, status	Daily festival volunteer shift logs (PRESENT, ABSENT, LATE, EXCUSED).

audit_logs	id (Serial)	user_id	action, target_type, target_id	Immutable platform audit trail tracking administrative actions and state diffs.
checkin_logs	id (Serial)	None	registration_id	Audit log recording gate check-in attempts, coordinator agent, and notes.
password_reset_tokens	id (Serial)	user_id -> users.id	token_hash (UQ)	Time-delimited SHA-256 password recovery tokens with automated expiration.
activities	id (Serial)	None	category	Fest feed featuring highlights, announcements, and cultural activity cards.
gallery_items	id (Serial)	None	None	Festival photo showcase with captions in Kannada and English.

5. Role-Based Access Control & User Journeys

Authentication is enforced via stateless JSON Web Tokens (JWT) signed with HS256. User roles determine route eligibility:

- **SUPERADMIN:** Master authority. Unrestricted dashboard access, live metric counters, volunteer attendance controls, Excel exports, admin approval lifecycle, and database inspection.
- **ADMIN (Event Coordinator):** Festival operations. Event attendee lists, real-time camera QR scanner for gate check-in, attendance viewing, and participant verification.
- **VOLUNTEER:** Event operations staff. Attendance logging, duty check-in, spectator assistance, and participant support.
- **PARTICIPANT / STUDENT:** Registered attendees. Event enrollment, solo/team registration, live schedule tracker, password change, and digital QR pass generation.
- **SPECTATOR / PUBLIC:** Open access. Cultural showcase exploration, Kannada literary schedule, event rules, gallery, and contact information.

6. Key Engineering Innovations & Problem Resolutions

- **Zero-Persistence Ephemeral Storage Fix:** Vercel serverless functions recycle containers on idle. The previous architecture stored data in `/tmp/akv_fest.db`, causing registered accounts to vanish. This was solved by engineering a hard configuration guard that rejects SQLite in production and connects exclusively to persistent Supabase PostgreSQL.
- **Serverless Connection Pooling:** Relational database connections are optimized for stateless Lambda functions using Supabase's transaction pooler (port 6543) combined with SQLAlchemy `pool_pre_ping=True` and `pool_recycle=300`.
- **Cross-Environment Reverse Proxy:** Configured Vite's local dev server to transparently proxy `/api` calls to `http://127.0.0.1:8000`, ensuring the exact same relative API URLs function in both local development and Vercel production without CORS conflicts.
- **Dual Dialect Schema Migrations:** Database bootstrap scripts automatically detect dialect (SQLite vs PostgreSQL) and query `information_schema.columns` or `PRAGMA table_info` to ensure schema alignment without destructive resets.
- **PostgreSQL Sequence Realignment:** A dedicated migration script (`scripts/migrate_sqlite_to_pg.py`) transfers all records while updating underlying PostgreSQL SERIAL sequences (`setval`) to avoid primary key collisions on

subsequent registrations.

7. Production Verification & Test Results

The live production deployment (<https://akv-nuditaranga-2026.vercel.app>) was validated using an automated end-to-end integration test suite verifying the complete registration and persistence lifecycle:

Test Case	Input / Action	Expected Result	Production Status
1. Event Catalog	GET /api/events	Return bilingual cultural competitions	PASSED (200 OK)
2. Superadmin Auth	POST /api/auth/login/admin	Issue JWT token with role claims	PASSED (200 OK)
3. Student Registration	POST /api/auth/register/student	Persist user record in Supabase PG	PASSED (201 Created)
4. Session Logout	Client token purge	Session cleared from browser	PASSED (Verified)
5. Re-Login with Same Credentials	POST /api/auth/login/student	Authenticate against PostgreSQL user	PASSED (200 OK)
6. Database Persistence Audit	Direct SQL query to Supabase	Record exists across serverless cold starts	PASSED (Confirmed)
7. Invalid Password Guard	POST /api/auth/login/student (wrong pw)	Reject with HTTP 401 Unauthorized	PASSED (401 Verified)

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